

Understanding Pointwise vs. Uniform Convergence of Functions

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I. FUNDAMENTALS

Before defining the core convergence topics of the paper, we must cover some ground rule definitions:

A. Normed Vector Spaces

Let E be a vector space. A *norm* on E is a function: $v \rightarrow |v|$ from E into $\mathbb{R}$ that satisfies the following axioms:

1. $|v| \geq 0$, and $|v| = 0 \iff v = 0$.
2. $\forall A \in \mathbb{R}, v \in E, |av| = |a||v|$.
3. $\forall v, w \in E, |v + w| \leq |v| + |w|$.

A vector space with a norm is called a normed vector space [3].

B. Supremum

For a set S , we say the set is bounded from above if there is a number c such that: $\forall x \in S, x \leq c$. We refer to c as an upper bound of S . Let R be the set of all upper bounds of S . We then define the least upper bound b as the minimum element of R . The element b is frequently called $\sup(S)$ [3].

C. Sup Norm

Let f be a bounded function on S . We define the sup norm of f , $\|f\| = \sup_{x \in S} |f(x)|$.

Observe that:

1. $\|f\| = 0 \implies |f(x)| = 0$ for all x . Also, $|f(x) = 0| \implies \sup_{x \in S} |f(x)| = 0$.
2. consider f and g . $\|fg\| = \sup_{x \in S} |g(x)f(x)| = \sup_{x \in S} |g(x)| \sup_{x \in S} |f(x)| = \|g\|\|f\|$.
3. Let $\|f\| = M$, $\|g\| = N$. Then $\|f + g\| = \sup_{x \in S} |f + g| \leq \sup_{x \in S} |f| + \sup_{x \in S} |g| = \|f\| + \|g\|$.

Thus, the sup norm is a norm.

II. POINTWISE CONVERGENCE

Pointwise convergence is the first form of convergence we will discuss. This form of convergence takes a sequence of mappings $\{f_n(x)\}$. If we can find an N value after which all subsequent $f_n(x) : n \geq N$ are in an epsilon neighborhood V_ϵ , then we say the sequence $f_n(x)$ converges pointwise. Note that the N in this definition is specific to the x we are looking at. Below is the formal definition.

A. Definition

Let S be a set, and F a normed vector space. Let $\{f_n\}$ be a sequence of maps: $S \rightarrow F$. For each $x \in S$ we may then consider the sequence of elements of F given by $\{f_1(x), f_2(x), \dots\}$. Thus for each x , we may speak of the convergence of the sequence $\{f_n(x)\}$. If $\{f_n\}$ is a sequence of maps such that for each $x \in S$ the sequence $\{f_n(x)\}$ converges, then we say that $\{f_n\}$ converges pointwise [3].

B. Examples

One example we can consider is the $\lim_{x \rightarrow \infty} \frac{1}{n}x^2 = 0$. This limit converges pointwise since for $\epsilon > 0, \epsilon \in \mathbb{R}$ there exists an $N = \frac{x^2}{\epsilon}$ such that $\forall n > N$ we know $|\frac{1}{n}x^2| < |\frac{1}{\frac{x^2}{\epsilon}}x^2|$ as $n > \frac{x^2}{\epsilon}$. Thus $|\frac{1}{n}| < \frac{1}{\frac{x^2}{\epsilon}}x^2 = \frac{\epsilon}{x^2}x^2 = \epsilon$ for all x . Notice our N value is dependent on the x we choose. Informally, this is equivalent to saying that different parts of the function converge to 0 at different rates.

Another example is the sequence of functions that look as follows:

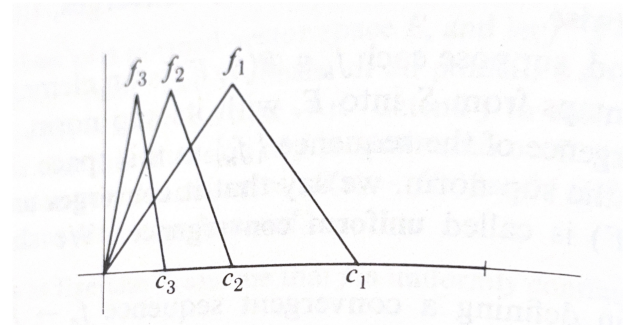


FIG. 1: credit: [3]

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Where we define $f_n(x) = 0$ when $x \geq c_n$. The limit of the functions will always equal zero, regardless of x , even if the convergence has differing rates depending on the x we choose. Thus the sequence $\{f_n\}$ converges pointwise to zero [3].

III. UNIFORM CONVERGENCE

Next is uniform convergence. This form of convergence does not depend on x . Thus, for something to uniformly converge, we consider the most extreme case (the sup norm), and if this value converges, all other cases must also converge. In other words, we find an upper bound for our mapping such that irrespective of x our mapping converges to a function f in our target space.

A. Definition

Let S be a set, and F a normed vector space. Let $\{f_n\}$ be a sequence of maps: $S \rightarrow F$. For each $x \in S$ we may then consider the sequence of elements of F given by $\{f_1(x), f_2(x), \dots\}$. Suppose each f_n is an element of the vector space of bounded maps: $f_n \in \beta(S, F)$ from S into F , with its sup norm. Then we may speak of the convergence of the sequence $\{f_n\}$ in this space. If the sequence $\{f_n\}$ converges for the sup norm, we say that it converges uniformly. Convergence in $\beta(S, F)$ is called uniform convergence. [3].

B. Examples

An example of uniform convergence is $\lim_{n \rightarrow \infty} \frac{1}{n} \sin x = 0$. For a given $\epsilon > 0, \epsilon \in \mathbb{R}$ observe that $\forall n > \frac{1}{\epsilon}$ we know $|\frac{1}{n} \sin x| \leq |\frac{1}{n}| = \frac{1}{n}$. Thus $|\frac{1}{n} \sin x| \leq \frac{1}{n} < \frac{1}{\epsilon} = \epsilon$. Observe that our N does not vary with respect to x , thus there is a uniformity to the rate of convergence of the function.

Returning to this example:

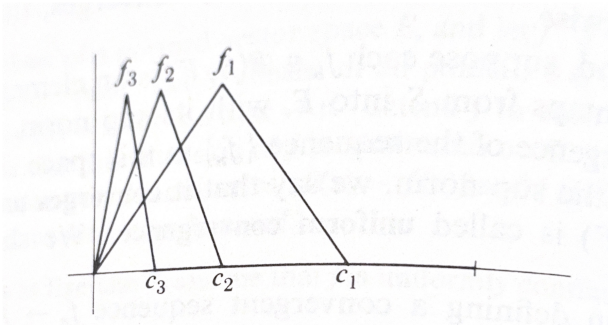


FIG. 2: credit: [3]

If we let all peaks of f_n go to one, then the sup norm

causes the pointwise converging sequence to not be uniformly convergent:

$$\|f_n\| = 1 = \|f_n - 0\|$$

Hence, this sequence is pointwise but not uniformly convergent [3].

IV. COMPARING POINTWISE AND UNIFORM CONVERGENCE

Observe first off that any function which converges uniformly by definition converges absolutely. This can be verified quickly by recognizing that the choice of N which makes the $\|f(x)\|$ converge is clearly one which will make every element of $f(x)$ converge pointwise.

V. PREFERENCE OF UNIFORM CONVERGENCE WHEN PRESERVING PROPERTIES LIKE CONTINUITY OR INTEGRABILITY

There are two major theorems in regards to uniform convergence. First we have the uniform limit theorem [2].

Theorem 1 *Let $\{f_n\}$ be a sequence of mappings from a metric space M to a metric space N . If $\forall n \in \mathbb{N} : f_n$ is continuous at every point of M and $\{f_n\}$ uniformly converges to f , then f is continuous at every point of M*

An example of a limiting sequence of continuous functions which does not converge to a continuous function can be found in Figure 4. Clearly the sequence in the picture converges to $\begin{cases} 1 & x = \frac{\pi}{2} \\ 0 & x \neq \frac{\pi}{2} \end{cases}$ which is non-continuous. However, $\sin^n(x)$ is continuous for all n .

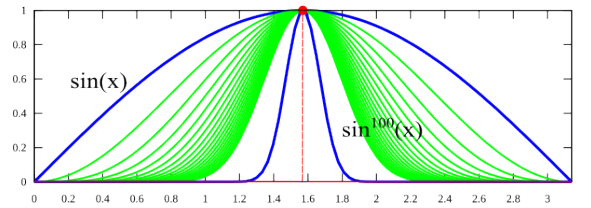


FIG. 3: Continuous Function Sequence [1]

The second theorem has to do with the Riemann integral [4].

Theorem 2 *If $\{f_n\}$ is a sequence of functions which uniformly converge to f then $\lim_{n \rightarrow \infty} \int_a^b f_n = \int_a^b f$. Where $\int$ is the Riemann integral.*

This interchange is not possible for pointwise convergence, as can be seen with the function sequence

$$f_n(x) = \begin{cases} n^2x & 0 \leq x < \frac{1}{n} \\ 2n - n^2x & \frac{1}{n} \leq x < \frac{2}{n} \\ 0 & x < 0, \frac{2}{n} \leq x \end{cases}.$$

Figure 4 is an image of this sequence. Observe that the area of the triangles is consistently 1 because they have height n and width $\frac{2}{n}$. This means that the integral of each of the functions is 1. Thus $\lim_{n \rightarrow \infty} \int f_n(x) = \lim_{n \rightarrow \infty} 1 = 1$. Taking the limit of the function sequence clearly yields the zero function. We know that the integral of the zero function is zero, this means that $\int \lim_{n \rightarrow \infty} f_n(x) \neq \lim_{n \rightarrow \infty} \int f_n(x)$.

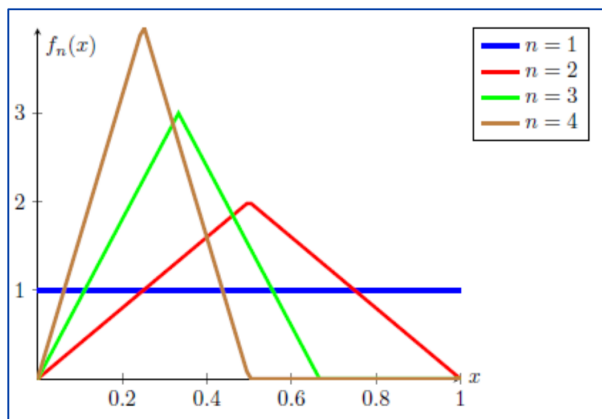


FIG. 4: Integration of a Function Sequence [4]

VI. CONCLUSION

In conclusion, uniform convergence is a much stronger condition for the convergence of a function sequence. It implies both point-wise convergence and implies certain conditions about integral interchange and continuity.

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